

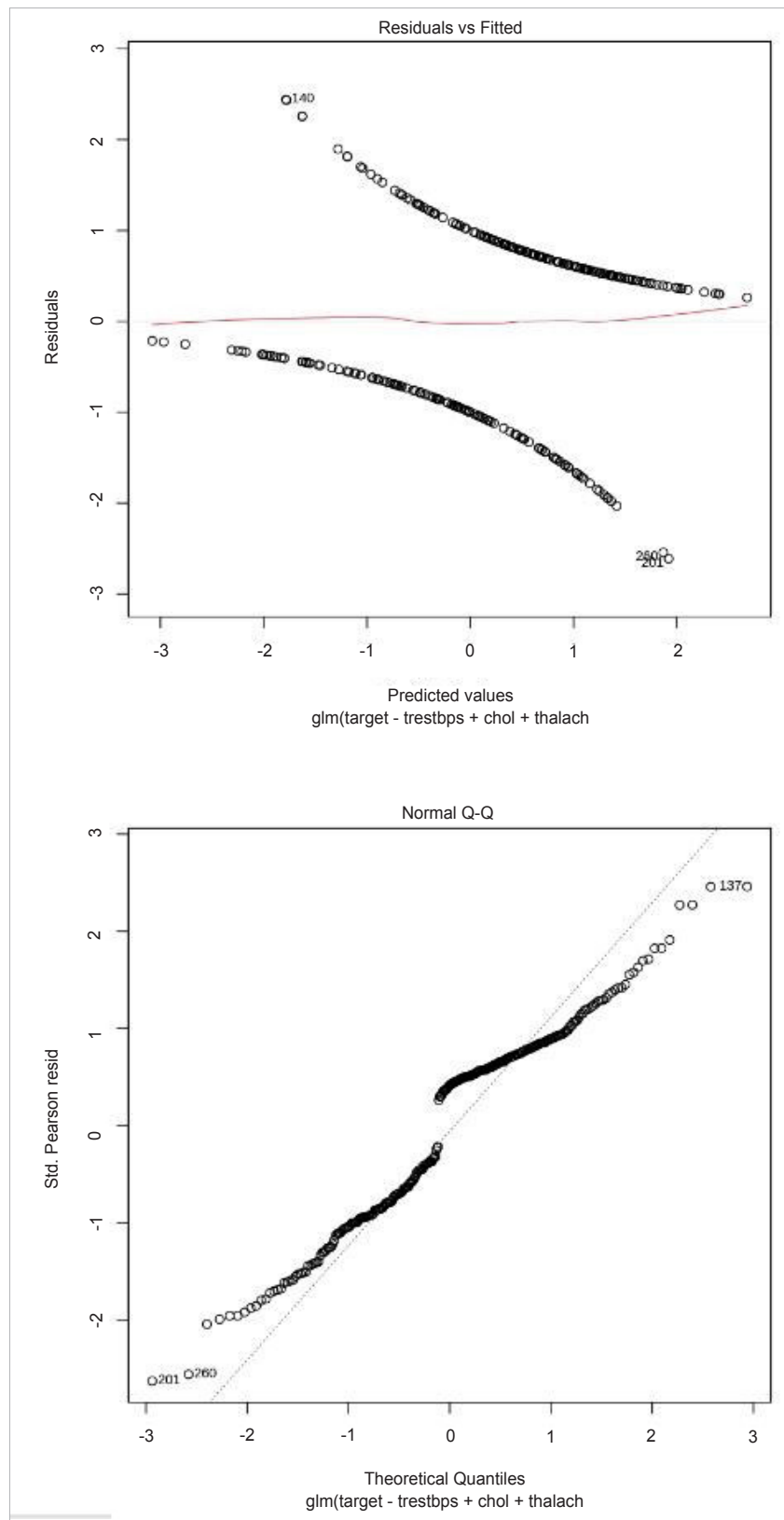


Figure 7: Graphical representation of residuals vs filtered

```
print("Construction of the Decision
Tree Model") mtree <- rpart(target
~ trestbps + chol + thalach, data
= train, method="class", control
= rpart.control(minsplit = 20,
minbucket = 7, maxdepth = 10,
usesurrogate = 2, xval =10))
```

Now plot the decision tree for the dataset, using 'mtree':

```
print("Plotting the Decision Tree")
plot(mtree)
text(mtree)
par(xpd = NA, mar = rep(0.7, 4))
plot(mtree, compress = TRUE)
text(mtree, cex = 0.7, use.n =
TRUE, fancy = FALSE, all = TRUE)
prp(mtree, faclen = 0,box.palette =
"Reds", cex = 0.8, extra = 1)
```

Next, construct the random forest model:

```
rf <- randomForest(target ~ trestbps
+ oldpeak + cp, data = data)
```

View the forest results:

```
print("Random Forest Results:")
print(rf)
```

The importance of each predictor can be gauged by:

```
print("Importance of each
predictor:")
print(importance(rf,type = 2))
```

To plot the random forest, type:

```
plot(rf)
```

The graphical outcomes of the regression model are depicted in Figures 7 and 8.

This IoT-based healthcare application for predicting the possibility of a heart attack using GLMs, random forests, and decision trees in R holds significant promise for predictive